

Smart University Cognitive Data Platform

Technical Architecture & Real Code Implementation Specification

1. Executive System Architecture & Topology

The Smart University Cognitive Data Platform is an enterprise-grade AI system that unites unstructured academic policies with transactional university data (student transcripts, fee records, timetables, and exam seatings).

Core Architecture Stack: Next.js 15 (App Router with React 19) + Prisma ORM + PostgreSQL (with pgvector & tsvector) + Qdrant Vector Database + Neo4j Knowledge Graph + LangGraph Multi-Agent Orchestrator + Python Enterprise Tool Runtime (ETR v3.0.0 FastAPI microservice).

Architectural Invariant: Probabilistic Large Language Models (LLMs) are restricted strictly to language comprehension, semantic synthesis, and context explanation. All institutional decisions—including exam eligibility, hall ticket clearance, fee validation, faculty workload balancing, and timetable conflict resolution—are computed by deterministic, rule-based computational engines.

2. Cognitive LangGraph Multi-Agent Orchestration Pipeline

The reasoning runtime is built upon LangChain's LangGraph framework (src/ai/graph/workflow.ts) operating under a hard 2000ms latency budget with progressive fallbacks.

- Query Complexity Router (queryComplexityRouterNode):** Classifies queries into SIMPLE (greeting, identity, memory lookup) vs COMPLEX (multi-step RAG, analytics, operational tasks).
- Fast Path (memoryNode):** SIMPLE queries route immediately to memory, bypassing vector searches and planning (< 30ms). If output is blank, an automated fallback re-enters planning.
- Unified Planner Node (unifiedPlannerNode):** Executes single-pass goal recognition, intent categorization, and task decomposition, replacing legacy multi-turn analysis latency.
- Subsystem Agent Nodes:** Dispatches to knowledgeNode (Tri-Modal RAG), databaseNode (Prisma/SQL tools), workflowNode (Deterministic business logic), or documentDeliveryNode (Structured document delivery).
- Combined Merge Node (combinedMergeNode):** Parallel fan-out merging that synthesizes quantitative DB tables with qualitative RAG policy citations.
- Verification Budget Router:** At cumulative elapsed time > 1800ms, dynamically skips verificationNode straight to responseNode to guarantee sub-2000ms SLA compliance.

3. Tri-Modal Hybrid Retrieval (RAG Pipeline)

The retrieval pipeline (src/server/services/retrieval.service.ts) executes a 12-stage retrieval and evaluation cycle ensuring factual grounding and zero hallucinations:

Stage	Component & Implementation	Technical Mechanism
1. Session Cache	Deterministic Memory Cache	SHA-256 hash indexed by org, dept, role, and normalized query.
2. RBAC Filter	DocumentAccessPolicy	Pre-retrieval query scoping (UNIVERSITY, COLLEGE, DEPARTMENT, PRIVATE).
3. Query Expansion	QueryIntelligenceService	Pronoun resolution from history + 3 semantic query variants.
4. Vector Search	Qdrant Dense Retriever	1536-dimensional cosine similarity embeddings across document chunks.
5. BM25 Search	PostgreSQL tsvector	Native sparse keyword matching with weighted document column ranking.
6. Graph Traversal	Neo4j Knowledge Graph	Entity-relation Cypher queries (Course prerequisites, department chains).

7. Rank Fusion	Reciprocal Rank Fusion (RRF)	RRF score = $\text{sum}(w_m / (60 + \text{rank_m}))$ fusing dense, sparse, and graph results.
8. Reranking	Local WASM BGE Reranker	Cross-encoder model executed in-process via WebAssembly (sub-15ms).
9. Compression	Jaccard Compressor	Prunes redundant semantic chunks (Jaccard similarity > 0.75).
10. Multi-Doc Prompt	PromptService	Formats chunk provenance, regulation IDs, and page numbers.
11. Triad Eval	EvaluationService	Logs Recall, Faithfulness, and Hallucination scores to DB.
12. Trace Logging	RetrievalLogService	Records full latency breakdown, query variants, and candidate metadata.

4. Deterministic Institutional Operations & Decision Engines

- **Hall Ticket & Examination Eligibility Engine (hall-ticket-engine.ts):** Deterministically evaluates Attendance ($\geq 75.0\%$), fee balance (≤ 0.00), continuous assessment completion, and disciplinary status. Returns structured decisions with regulation citations and remedial steps.
- **Faculty Workload & Timetable Engine (workload-engine.ts, faculty-conflict-engine.ts):** Computes credit-hour teaching loads vs statutory limits (12/14/16 hrs by rank) and verifies 3D collision freedom across Rooms x Faculty x Sections over 6-day academic timetables.
- **Student Academic Risk Predictor (student-risk-predictor.ts):** Evaluates multi-semester SGPA trends, backlog counts, and attendance trajectories to assign a risk score [0.0 - 1.0] and trigger early interventions.
- **Cognitive Kernel & Policy Engine (policy-engine.ts):** Enforces regulatory policies: POL-001 (Exam attendance limits), POL-002 (Bulk communication sign-offs), and POL-003 (RBAC grade change constraints).

5. Relational Data Architecture (Prisma Schema)

- **Tenancy & Identity:** Organization, User, Membership, ApiKey, AuditLog.
- **Academic Hierarchy:** College, Department, Faculty, Course, CourseSection, Student, Enrolment, Facility, ResearchProject.
- **Student Operations:** AttendanceRecord, InternalMark, SemesterResult, HostelRecord, ScholarshipRecord, ParentInfo, FinancialAccount.
- **Examination Operations:** Examination, ExaminationSchedule, ExaminationEligibility, InvigilationAssignment, TimetableEntry, ExamSeatingArrangement.
- **AI Telemetry & Knowledge:** KnowledgeBase, Document, Chunk, RetrievalLog, QueryVariant, Evaluation, AgentRun, UsageEvent, PlanLimits.

6. Python Enterprise Tool Runtime (ETR v3.0.0 Microservice)

A dedicated FastAPI microservice (services/python-agent-service) runs on port 8000 providing an isolated Enterprise Tool Runtime (ETR). It features dynamic tool discovery, execution context scoping, Prometheus metric collection, circuit-breaker timeout wrappers, and RBAC/ABAC policy validation.

7. Institutional Portals & User Workspaces

- **HOD & Dean Command Center (/hod/*):** Real-time department health scores (Academic, Attendance, Research), 'What Changed' delta analytics, automated action proposal review, and syllabus comparison tool.
- **Faculty Workspace (/faculty/*):** Personal teaching schedule, automated invigilation duty rosters, hall seating arrangements, and authorized course document delivery.
- **Enterprise AI Studio (/dashboard/*):** Streaming conversational interface with live LangGraph execution trace, Retrieval Debugger with raw vector/BM25/RRF scoring breakdown, and token usage analytics.